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2041: The Next 20 Years



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Standing at the technological frontier we are at today; it is probably hard to imagine times when flatscreen TVs and wired telephones were a source of wonder. But to gauge where we will be 20 years into the future, we need to step back and evaluate how the past two decades have shaped our reality today. Around the turn of the millennium, the smartphone heralded a new technological revolution. The early smartphones (Palm pilot, communicator, etc.) were quite complicated to use. The multiple design and OS iterations that the smartphone has undergone since then have shown us the pathways to the methodology of technological evolution. It is the best example to learn how innovations of today can become the reality of tomorrow.

The benefits of any new technology are initially outweighed by certain key factors such as usability and cost barriers. As the accessibility of technology increases over time with innovation, the perceived value of the device rises

for the consumer. Large-scale adoption of technology is the route to making it a part of our reality. To facilitate this adoption, advancements in technology make the user experience more optimised. This increases the value proposition of the product, making it more popular. As the popularity and usage of the product increase, the cost eventually reduces.

Electric vehicles (EVs) for example, are in the phase of crossing the usability and cost barrier. EVs were written off in their earlier iterations as they were prohibitively expensive and impractical because of battery, performance, charging infrastructure and other issues. But technological innovations have made great strides since then. This has ensured that factors such as battery costs have come down. The range has climbed considerably and charging stations and solutions are becoming more commonplace, making it a practical option.

The Next 20 years: Life Transformation Through Democratisation of Data

In the recent past, we have seen an increase in the amount of data available to us. The exponential rise in that figure over the next 20 years can only be imagined along with the connectivity we will need to process it. As we all navigate this hyperconnected sea of data, information density increases, multi-tasking increases and new interaction paradigms come into play. This new interaction paradigm holds the potential to change our cognitive thought process and life experiences. Two technologies will aim to simplify this shift over the next two decades:

Natural Language processing (NLP)

Voice is the most natural form of communication for humans. NLP will provide the most effective way of human machine interface. Today, we already have an established interface with a host of digital assistants. NLP,

in its developmental stage at present, combines AI and Machine Learning to process human commands for completing assigned tasks. It holds the potential to revolutionize our current interaction with technology. Going forward, it will be the primary interface for smart homes, automobiles, healthcare, governance, and business solutions to substantially enhance our quality of living and productivity.

At present, customer experience analysis focuses on statistical data supplied by web crawlers. An emotional response is something we do not expect from analytical tools available in our time. However, NLP is also making new inroads in sentiment analysis, which will go the extra mile in taking business decisions based on the emotional response in customer experience.

Immersive technologies

AI is redefining all dimensions of our digital reality. However, there is a need for a connecting tech that will allow everyone to enjoy the benefits of AI. Immersive technologies like VR, AR & MR will be the connective tissue that will make AI more commonplace. AR has proven its utility in education and professional training since it found its feet in the gaming industry. Modules designed with AR have reduced the risks in medical procedures, making practitioners more accurate. Real-life testing is adding to the viability of this technology as it holds untold potential in the field of interactive learning.

This leap in technology will also be vital in business applications such as unification of global payment systems and solutions for logistics on a global scale. In the future, this knowledge can be applied for core design for products to simplify lives.

In conclusion, technologies such as NLP and immersive technologies like AR, VR & MR are indicative of bringing about a sea of change in our way of life over the next 2 decades. The future does hold the same wonder for us, as the smartphone did for the millennium. It would be interesting to see whether we will all have our own personal assistants in the form of a Jarvis or C3PO in the next 20 years. **d**